INCH-POUND

MIL-B-2316E
31 December 1992
SUPERSEDING
MIL-B-2316D
29 May 1973

MILITARY SPECIFICATION

BAKERY EQUIPMENT, INCLUDING UNIT ASSEMBLIES

REPAIR PARTS AND TOOLS, PREPARATION FOR

DELIVERY OF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the packaging, packing, and marking for shipment of bakery equipment and related items designated for use in garrison bread bakeries.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5018 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

FEDERAL

TT-P-664	-	Primer Coating, Alkyd, Corrosion-Inhibiting, Lead and Chromate Free, Voc-Compliant
PPP-B-576	-	Box, Wood-Cleated, Panelboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-1055	-	Barrier Material, Waterproofed, Flexible
PPP-B-1163	-	Box, Corrugated Fiberboard, High Compression Strength, Weather-Resistant, Wax-Resin Impregnated
PPP-B-1364	-	Boxes, Shipping, Corrugated Fiberboard, High Strength, Weather Resistant, Double Wall
PPP-P-40	-	Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Wood Working Machinery
PPP-P-600	-	Porcelain Enamel Products and Household Appliances, Electrical and Mechanical - Requirements for Packaging of
PPP-T-60	-	Tape: Packaging, Waterproof

MILITARY

MIL-V-3	-	Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of
MIL-C-104	-	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	-	Preservation, Methods of
MIL-P-551	-	Pipes and Elbows, Air Conditioning - Heating
MIL-R-3593	-	Refrigeration and Cooling Equipment (Excluding Household Refrigerators, Packaging of
MIL-C-3774	-	Crates, Wood; Open 12,000- and 16,000-Pound Capacity
MIL-C-52950	-	Crates, Wood, Open and Covered
MIL-E-16298	-	Electric Machines Having Rotating Parts, Accessories and Associated Support Items: Packaging of

STANDARDS

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1186	-	Cushioning, Anchoring, Bracing, Blocking and Waterproofing; With Appropriate Test Methods

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3950 - Standard Specification for Strapping, Plastic (and Seals)
D 3951 - Standard Practice for Commercial Packaging
D 4727 - Standard Specification for Corrugated and Solid Fiberboard Sheet Stock (Container Grade) and Cut Shapes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

3.1.1 Level A.

3.1.1.1 Disassembly. The extent of disassembly shall be determined by the manufacturer. This determination shall be based on the need for protection of parts susceptible to damage during transit or storage and for reduction in cubage. Attaching hardware shall be replaced on one of the mating parts or placed in a cloth drawstring bag and the bag attached to one of the mating parts. Parts removed shall be adequately identified or tagged. Where disassembly affects the operation of the equipment, warning tags or labels indicating the disassembly shall be attached. Instructions for assembly shall be included either as a part of the warning tag, or label, or as a separate sheet or folder attached to the equipment. Disassembly shall not be such that special skills or special tools, other than those furnished, or both, are required to place equipment in operation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

3.1.1.2 Draining. All water or fuel containing parts, such as water cooling systems, tanks, burners, and hoses, shall be completely drained and thoroughly dried.

3.1.1.3 Cleaning. All exposed, uncoated machined surfaces and critical operating surfaces shall be cleaned according to process C-5 and all other exposed, uncoated, ferrous metal surfaces shall be cleaned according to process C-1, and thoroughly dried in accordance with MIL-P-116.

3.1.1.4 Repainting and recoating. Painted and permanently coated surfaces from which the paint and permanent coating have been removed or on which the finish is defective shall be repainted and recoated with material identical to the original.

3.1.1.5 Lubrication. All bearings and fittings requiring lubrication shall be lubricated to capacity with the manufacturer's recommended lubricant containing a rust inhibitor.

3.1.1.6 Gasoline tanks. After draining and drying, the interior of the gasoline tanks shall be fogged with type P-10 preservative of MIL-P-116 and then pressure fogged with type P-18 preservative in accordance with MIL-P-116. The cap shall be replaced and the breather hole sealed with tape conforming to PPP-T-60.

3.1.1.7 Burners, combustion chamber, and radiators. Burners, combustion chamber, and radiators shall be coated with type P-7 preservative of MIL-P-116.

3.1.1.8 Pulleys. Drive belts shall be removed and exposed, uncoated ferrous metal surfaces of pulleys and shafts shall be coated with primer conforming to TT-P-664, and thoroughly dried. The belts shall then be replaced and tension relieved.

3.1.1.9 General preservation. Preservation of equipment and related parts not specified shall be as follows: All cleaned contacting surfaces (surfaces that contact other surfaces in operation), except food contacting surfaces, shall be coated with type P-2 preservative of MIL-P-116. All cleaned noncontacting surfaces (surfaces that do not contact other surfaces in operation), except food contacting surfaces, shall be coated with type P-1 preservative of MIL-P-116. All cleaned, uncoated ferrous metal surfaces which will be in contact with food ingredients shall be coated with type P-14 preservative of MIL-P-116.

3.1.1.9.1 Protection of preservative. Coated surfaces, from which preservative might be removed by contact with packaging and packing materials, shall be covered, wrapped or bagged with greaseproof barrier material in accordance with method I of MIL-P-116.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

3.1.1.10 Electrical motors. All open type motors not exceeding 40 pounds in weight, disassembled from the basic unit, shall be preserved in accordance with method II of MIL-P-116, with selection of the submethod at the option of the manufacturer. Totally enclosed motors shall not require any preservation other than that specified in 3.1.1.5.

3.1.1.11 Technical literature. Technical literature shall be packaged in accordance with method IC-1 of MIL-P-116, and secured to the parent unit.

3.1.1.12 Sealing. Lamps, electric cable connections, receptacles, switches, terminals, openings into motors, instruction and identification plates, control boxes, gages, meters, locks, key openings, and crevices formed by covers or lids shall be completely sealed with tape conforming to PPP-T-60. Remaining openings into the units shall also be sealed with the same type tape as specified herein.

3.1.1.13 Refrigeration and cooling systems. Refrigeration and cooling systems shall be processed in accordance with MIL-R-3593.

3.1.1.14 Miscellaneous items and accessories. Miscellaneous items and accessories shall be preserved and packaged in accordance with the applicable requirements of the following specifications:

Electrical equipment	MIL-E-16298
Hand tools	PPP-P-40
Pipe and elbows	MIL-P-551
Valves, fittings, and flanges	MIL-V-3

3.1.1.15 Securing. Parts of the equipment that are free to move shall be secured in a fixed position to prevent movement or vibration while in transit.

3.1.1.16 Disassembled components, tools, and accessories. Disassembled components, tools, and accessories shall be packaged in a fiberboard or nailed wood box conforming to style optional, type CF or SF class domestic of PPP-B-636, or class 1, style 2 or 4 of PPP-B-621, respectively. The weight of contents shall not exceed the weight limitations of the applicable box specification. Contents shall be cushioned, blocked, or braced within the box to prevent movement while in transit. Each box shall be securely closed in accordance with the appendix of the applicable box specification. A disassembled component considered too large or heavy to package may be packed directly into a shipping container or with its parent item.

3.1.2 Level C. Bakery equipment, units, assemblies, repair parts and tools shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from supply source to first receiving activity in accordance with ASTM D 3951. The supplier may use his standard practice when it meets these requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

3.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

3.2.1 Level A.

3.2.1.1 Bakery equipment not exceeding 65 pounds. Bakery equipment, units, and assemblies not exceeding 65 pounds, preserved and packaged as specified in 3.1.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style optional, V2s of PPP-B-636. Fiberboard pads or forms conforming to ASTM D 4727 shall be used, as required, to prevent movement or damage of the contents during transit. Each shipping container shall be closed, waterproofed, and reinforced with flat steel strapping or tape banding in accordance with PPP-B-636. Bakery equipment, units, and assemblies weighing 40 pounds or less, preserved and packaged as specified in 3.1.1, shall be packed in containers conforming to style optional, V3s of PPP-B-636. Cushioning of contents, and closure, waterproofing, and reinforcing of shipping containers shall also be as specified in PPP-B-636.

3.2.1.2 Bakery equipment not exceeding 1,000 pounds, except as specified in 3.2.1.4. Bakery equipment, units, and assemblies not exceeding 1,000 pounds preserved and packaged as specified in 3.1.1, shall be packed in wood-cleated plywood or nailed wood shipping containers conforming to overseas type of PPP-B-601 or class 2, style 2, 4, or 7, type 3 load of PPP-B-621, respectively. Alternately, a shipping container as specified in 3.2.1.3 may be used at the option of the contractor for equipment, units, and assemblies weighing from 500 to 1,000 pounds. Each shipping container exceeding 200 pounds gross weight shall be provided with skids as specified in the applicable container specification. Contents shall be secured as specified in 3.2.1.5. The top, sides, and ends of the nailed wood shipping container conforming to style 7 of PPP-B-621 shall be lined with waterproof barrier material conforming to PPP-B-1055, with all joints having a 4 inch minimum single lap. The contents of each shipping container conforming to style 2 or 4 of PPP-B-621 shall be waterproofed in accordance with the appendix of MIL-C-52950. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification.

3.2.1.3 Bakery equipment exceeding 1,000 pounds, except as specified in 3.2.1.4. Bakery equipment, units, and assemblies exceeding 1,000 pounds, preserved and packaged as specified in 3.1.1, shall be packed in a sheathed, nailed wood crate conforming to type I, class 1 or 2, style A of MIL-C-104. Contents shall be secured as specified in 3.2.1.5.

3.2.1.4 Troughs, racks, cabinets, and stands.

3.2.1.4.1 Weight not exceeding 200 pounds. Troughs, racks, cabinets, and stands, not exceeding 200 pounds, preserved and packaged as specified in 3.1.1, shall be packed in a snug-fitting, wood-cleated fiberboard or veneer overlaid shipping container conforming to type I or III, class 2, Style A or

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

B of PPP-B-576. Equipment having nondetachable casters or wheels shall be blocked in such a manner as to have a minimum clearance of 1 inch between the interior base or end of the container and the casters or wheels.

3.2.1.4.2 Weight not exceeding 2,500 pounds. Troughs, racks, cabinets, and stands not exceeding 2,500 pounds, preserved and packaged as specified in 3.1.1, shall be packed in an open wood crate conforming to type I, II, IV, or V, style A of MIL-C-52950. Each shipping container shall be closed, strapped, and waterproofed in accordance with the appendix of the container specification. Contents shall be secured as specified in 3.2.1.5.

3.2.1.5 Securing. Equipment provided with bolt holes for anchoring shall be secured to the base of the shipping container by bolting through the skids. When bolt holes and the skids are not in alignment, additional skids, longitudinal sleepers, or interior load bearing members shall be provided. The minimum size of the sleepers, or interior load bearing members shall be nominal 2 by 4 inch lumber 2 feet long. Blocking and bracing shall be provided as required to prevent sway of topheavy items. When bolt holes in the equipment being packed are not available, the equipment shall be blocked and braced in a manner to prevent movement and damage while in transit. Equipment having nondetachable casters or wheels shall be blocked in such a manner as to have a minimum clearance of 1 inch between the interior base or end of the container and the casters or wheels. A clearance of not less than 1 inch shall be provided between the items and the closest member of the sides, ends, and top of the shipping container. Cushioning and padding shall be used where necessary to prevent damage between the lumber blocking and unprotected finished surfaces. Wood blocks and braces shall be securely nailed to floorboards, headers, or other frame members when present. The blocking and bracing shall not depend on end-grain nailing alone but supplemental supporting cleats shall be provided to reinforce and prevent splitting of the blocking and bracing.

3.2.1.6 Storage of boxed components. Boxes containing items to be shipped with a unit or assembly shall be securely fastened in an unused portion within the shipping container of that unit or assembly.

3.2.1.7 Alternate Level A Packing. Alternatively, at the option of the supplier, the bakery equipment shall be packaged in accordance with the level A requirements and shall be packed to meet the test requirements for level A of PPP-P-600.

3.2.2 Level B.

3.2.2.1 Bakery equipment not exceeding 160 pounds. Bakery equipment, units, and assemblies not exceeding 160 pounds, preserved and packaged as specified in 3.1.1, shall be packed, unless otherwise specified (see 6.2), in a snug-fitting fiberboard shipping container conforming to style optional, type CF or SF, class domestic of PPP-B-636. When specified (see 6.2)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

fiberboard shipping containers conforming to style optional, V3c, V3s, or V4s of PPP-B-363, class 2, style G of PPP-B-640, SW or DW of PPP-B-1163 or PPP-B-1364 shall be used. Each shipping container shall be closed in accordance with the appendix of the applicable container specification. The weight of contents shall not exceed the weight limitations for the applicable grade of the specification of the container used. The contents of each container shall be immobilized in accordance with MIL-STD-1186.

3.2.2.2 Bakery equipment not exceeding 400 pounds. Bakery equipment, units, and assemblies not exceeding 400 pounds, preserved and packaged as specified in 3.1.1, shall be packed in a snug-fitting, wood-cleated fiberboard shipping container conforming to type I or III, class 1, style A or B of PPP-B-576. Each shipping container exceeding 200 pounds gross weight shall be provided with skids as specified in PPP-B-576 or provisions for handling with power equipment such as fork lift trucks. Other provisions for handling shall be by wood strips bolted to the base of the appliance and secured to the inside face of the bottom of the shipping container. These internal members shall be positioned so as to ensure protection to the container and contents when handling with power equipment. When internal strips are used, the container shall be marked to indicate two opposite sides intended for fork lift truck handling.

3.2.2.3 Bakery equipment not exceeding 1,000 pounds. Bakery equipment, units and assemblies not exceeding 1,000 pounds, preserved and packed as specified in 3.1.1, shall be packed in wood-cleated plywood or nailed wood shipping containers conforming to domestic type, style A or B of PPP-B-601 or class 1, style 2, 4, or 7, with diagonals as required, of PPP-B-621, respectively. Each shipping container shall be provided with skids as specified in the applicable container specification, and the contents secured as specified in 3.2.1.5. Alternatively, ovens weighing up to 750 lbs. net weight shall have the components packaged as specified in 3.1.1, placed within the oven and cushioned and blocked with fiberboard pads and forms. The oven shall be placed back down, doors up, on a sheet of cushioning material placed over a fully sheathed skid base. The base shall be provided with a full length and width retaining frame made from four pieces of wood, two minimum 1-3/4 x 1-3/4 inch and two minimum 3/4 x 1-3/4 inch securely nailed to the top deck of the base. The deck shall be nominal 1 inch lumber and skids nominal 2 x 4 inch lumber. The skids shall be positioned so as to support the box cover on four sides. The oven shall be secured to the base by four bolts or four bolted clips. Bolts, when clips are used, shall be minimum 3/8 inch carriage bolts long enough to go through the skid or 2 x 4 inch sleeper, deck boards and 1-3/4 x 3/4 inch retainer frame and through the clip. Bolts, when used to bolt to the frame of the unit shall pass through the skid or 2 x 4 inch sleeper, deck and the oven frame. Each bolt shall be secured with a lock-washer and a nut. The clip shall engage the frame, 2 each side and a minimum of 11 gage thick and 2 inches wide. The oven doors shall be secured by 2 bands of 5/8 x .020 inch steel strap encircling the oven and wood base. Edge protectors shall be placed between the strap and oven. The oven shall be completely covered by a full height shroud made from

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

minimum 0.004 inch thick polyethylene or material conforming to class E-2, E-1, H-5 or M-1 of PPP-B-1055. The box cover shall be an inverted 1/2 slotted HSC triple wall box conforming to style E, class 2 of PPP-B-640. The box cover shall be placed over the oven onto the base skids. The box cover shall be secured to the retaining frame with not less than 5 staples each side. The staples shall have a crown length of not less than 1-1/4 inches. The box cover shall then be strapped with two bands of nonmetallic strapping encircling the box cover and base. The strapping shall have a minimum break strength of 790 lbs. and be in accordance with type II or III of ASTM D 3950. There shall be a minimum of 1 inch clearance between the interior of the top sides and ends of the container and the oven.

3.2.2.4 Bakery equipment not exceeding 4,000 pounds. Bakery equipment, units, and assemblies not exceeding 4,000 pounds, preserved and packaged as specified in 3.1.1, shall be packed in a wood crate conforming to style A, type IV or style B, type I, II, or V of MIL-C-52950. Contents shall be secured as specified in 3.2.1.5.

3.2.2.5 Bakery equipment weighing from 4,000 to 12,000 pounds. Bakery equipment, units, and assemblies weighing from 4,000 to 12,000 pounds, preserved and packaged as specified in 3.1.1, shall be packed in a wood crate conforming to type I, style A of MIL-C-3774. Contents shall be secured as specified in 3.2.1.5.

3.2.2.6 Bakery equipment exceeding 12,000 pounds. Bakery equipment, units and assemblies exceeding 12,000 pounds, preserved and packaged as specified in 3.1.1, shall be packed in a sheathed, nailed wood crate conforming to type I, class 1 or 2, style A of MIL-C-104, except that waterproof barrier material shall not be required. Contents shall be secured as specified in 3.2.1.5.

3.2.2.7 Stowage of boxed components. Stowage of boxed components shall be as specified in 3.2.1.6.

3.2.2.8 Alternate Level B. Alternatively, at the option of the supplier, the bakery equipment shall be packaged in accordance with the level A packaging requirements and shall be packed to meet the test requirements for level B of PPP-P-600.

3.2.3 Level C. Bakery equipment, units, and assemblies, preserved and packaged as specified in 3.1.1, shall be packed in a manner to ensure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies in accordance with ASTM D 3951.

3.3 Marking. In addition to any special marking required by the contract or purchase order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129. Set marking requirements shall apply when accessories and parts are packed separate from the unit (see 6.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

3.3.1 Precautionary marking. Special handling marking requirements in MIL-STD-129 applicable to fragile items, such as arrows, the words "UP", "THIS SIDE UP" and "CENTER OF BALANCE" shall apply.

3.4 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document.

3.5 Workmanship. The end item shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be examined in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.2.2 Examination of the end item. The end item shall be examined for visual preparation of packaging, preservation and packaging, packing, and marking in accordance with the defects listed in tables I through IV. The lot size shall be expressed in terms of disassembled, packaged or packed bakery item equipment, as applicable. The sample unit shall be one

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

dissembled, packaged or packed item of bakery equipment, as applicable, of the specified type, design, size and characteristic. The inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

TABLE I. End item visual defects

Examine	Defect	Category Major
Disassembly (as applicable)	Detached parts or hardware not placed with mating parts as specified (see 3.1)	101
Matchmarking	Removed parts not identified or tagged (see 3.1.1.1)	102
	Warning tags or labels indicating the disassembly not attached or missing (see 3.1.1.1)	103
	Instructions for reassembly missing (see 3.1.1.1)	104
Draining (as applicable)	All water or fuel not removed from fuel or water containing parts or systems as applicable (see 3.1.1.2)	105
Repainting and recoating	Not repainted or recoated as needed (see 3.1.1.4)	106

TABLE II. End item preservation and packaging defects

Examine	Defects	Category Major
Lubrication (where applicable)	Not lubricated to capacity (see 3.1.1.5)	101
Gasoline tanks	Cap not replaced after fogging	102
	Breather hole not sealed with tape (see 3.1.1.6)	103
Pulleys and drive belts	Pulleys not preserved as specified	104
	Drive belts not replaced with tension relaxed (see 3.1.1.8)	105
Technical literature	Missing or not secured to parent unit (see 3.1.1.11)	106

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

TABLE II. End item preservation and packaging defects (cont'd)

Examine	Defects	Category Major
Sealing of units (as applicable)	Any openings or crevices not sealed with tape specified (see 3.1.1.12)	107
Movable parts of equipment	Not secured in a fixed position (see 3.1.1.15)	108
Disassembled components, tools and accessories	Not cushioned and securely immobilized within the box as specified (see 3.1.1.16)	109

TABLE III. End item packing defects

Examine	Defects	Category Major
Container	Not as specified for weight limits of equipment being packed (see 3.2.1.1 through 3.2.2.6 as applicable)	101
	Less than 1 inch clearance between item and closest member of sides, ends and top, when required (see 3.2.1.4, 3.2.1.5 and 3.2.2.3)	102
	Top, sides, and ends of nailed wood box (PPP-B-621) not lined with barrier as specified (see 3.2.1.2)	103
Skids or handling means (when required)	Missing (see 3.2.1.2 and 3.2.2.2)	104
	Strips not bolted to base of appliance (see 3.2.2.2)	105
	Strips not secured to bottom of container (see 3.2.2.2)	106
	Strips do not protect the unit or container from fork lift damage (see 3.2.2.2)	107
	Markings not as required for boxes with internal handling means (see 3.2.2.2)	108
Bakery equipment, units and assemblies	Not securely immobilized by bolting and blocking and bracing as specified (see 3.2.1.5)	109
	Longitudinal sleepers or load bearing members missing when required (see 3.2.1.5)	110
	Any wood member less than size specified (see 3.2.1.5)	111

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

TABLE III. End item packing defects (cont'd)

Examine	Defects	Category Major
Bakery equipment, units and assemblies (cont'd)	Bolting not positioned through the skids as specified (see 3.2.1.5)	112
	Cushioning and padding of finished surfaces not as specified (see 3.2.1.5)	113
	Nailing application of blocking and bracing not as specified (see 3.2.1.5)	114
Boxed components	Not secured within shipping container, when applicable (see 3.2.1.6)	115

TABLE IV. End item marking defects

Examine	Defects	Category Major
Marking interior packages and shipping container (as applicable)	Omitted; incorrect; illegible; or of improper size, location, sequence or method of application (see 3.3 and 3.3.1)	101
	Set markings missing, when applicable (see 3.3)	102
	Precautionary markings missing, or not as specified (see 3.3.1)	103

4.2.3 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-576, PPP-B-601, PPP-B-621, PPP-B-636, PPP-B-640, PPP-B-1163, PPP-B-1364, MIL-C-104, MIL-C-3774, or MIL-C-52950, examination for defects in closure, strapping, and water-proofing, when applicable, shall be in accordance with the applicable container specification.

5. PACKAGING

This section is not applicable to this specification.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The packaging, packing, and marking of bakery equipment and related items.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. Levels of packaging and packing required (see 3.1 and 3.2).
d. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 3.2.2.1).
e. When additional marking is required (see 3.3).

6.3 Subject term (key word) listing.

Burners
Draining
Electric
Gasoline
Motors
Packaging
Packing
Pullies
Refrigeration
Tank

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - YD
Air Force - 99

Preparing activity:

Army - GL

(Project 7310-0814)

Review activities:

Navy - MC
Air Force - 69, 84
DLA - GS

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-B-2316E	1992 December 31

3. DOCUMENT TITLE

BAKERY EQUIPMENT, INCLUDING UNIT ASSEMBLIES REPAIR PARTS AND TOOLS, PREPARATION FOR DELIVERY O

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON/DSN
U.S. Army Natick RD&E Center	508-651-4501	256-4501
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WTP Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T14:42Z
Check the source to verify that this is the current version before use.